

Lorenzo Grassi

Curriculum Vitae

* Born: Italy, 3 October 1989

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Updated: December 2025

Current Position

- From 01/02/2025 **Assistant Professor in (Symmetric) Cryptography (80% of the time)**
Eindhoven University of Technology (TU/e), Eindhoven, the Netherlands
(Department of Mathematics and Computer Science)
- Analysis and design of symmetric cryptographic primitives over prime fields and integer rings for modern applications
 - *Group Leader:* Prof. Tanja Lange (Coding Theory and Cryptology Group)
- From 01/04/2023 **Consultant at Ponos Technology – Cryptographer (20% of the time)**
- Analysis and design of Zero-Knowledge friendly symmetric cryptographic primitives targeting efficient Hardware implementation

Impact

ORCID: <https://orcid.org/0000-0003-1140-0520>

(<https://scholar.google.com/citations?user=bbb0VYMAAAAJ>)

No. publications: **48**

No. citations: **3230** (Google Scholar – 16/12/2025)

H-index: **25** (Google Scholar – 16/06/2025)

Education

- From 01/09/2015 **PhD in Symmetric Cryptography**
to 26/04/2019 *IAIK, Graz University of Technology, Graz (Austria)*
- *Thesis Title:* “Cryptanalysis of AES-like Ciphers and Reviving Old Design Ideas for New Constructions”
 - *Supervisor:* Prof. Christian Rechberger
- From 10/2011 to **Master Degree’s in Mathematics**
27/02/2014 *University of the Study of Milano (Italy)*
- *Final grades:* 110/110 cum laude (i.e., highest possible grade in Italian system)
- From 09/2008 to **Bachelor’s degree in Mathematics for Applications**
10/2011 *University of the Study of Milano (Italy)*

Research and Professional Experience

- From 15/03/2023 **PostDoc in Symmetric Cryptography (80% of the time)**
to 28/02/2025 *Ruhr University Bochum, Bochum, Germany*
- *Supervisor:* Prof. Gregor Leander
- From 01/03/2020 **PostDoc in Symmetric Cryptography**
to 28/02/2023 *Radboud University, Nijmegen (Netherlands)*
- *Supervisor:* Prof. Joan Daemen

From 01/02/2022 **Consultant at Ponos Technology – Cryptographer** (*Contract Hours*)
to 28/02/2023

From 27/04/2019 **PostDoc in Symmetric Cryptography**
to 29/02/2020 *IAIK, Graz University of Technology, Graz (Austria)*
○ *Supervisor:* Prof. Christian Rechberger

From 23/06/2014 **Internship at STMicroelectronics – Cryptographer**
to 22/12/2014 *Agrate Brianza (Italy)*
○ *Tutor:* Eng. Guido Marco Bertoni (AST Security Lab)

Research Grants

ERC 2024 “*SYMPZON – Getting SYMMetric CryPtography Out of its Comfort ZONE*”, European Research Council (ERC) Starting Grant, for 5 years starting in January 2025 – approx 1.500.000 euro

“*Design and Cryptanalysis of Symmetric Cryptographic Primitives for Zero-Knowledge Applications*”, Ethereum Foundation – Academic Grant Program 2024,¹ for 1 year starting in June 2024 – approx 50.000 euro

“*Design and Cryptanalysis of Symmetric Cryptographic Primitives for Secure Computations*”, CASA Jump.Start Program for Postdocs² (Germany), for 2 years starting in March 2023 – approx 200.000 euro

PhD Student(s)

From 01/10/2025 Berenika Richterová
Supported by the ERC starting grant 2024 “SYMPZON”

Service – Program Committees and Editorial Boards

FSE/ToSC 2025: **General Co-Chair** (together with Prof. Marco Pedicini)

Program Committee **EUROCRYPT 2026 – EUROCRYPT 2024**
Member: **CRYPTO 2026 – CRYPTO 2025 – CRYPTO 2024**
ASIACRYPT 2023

SCN 2026

LightSEC 2025

Africacrypt 2023 – Africacrypt 2022

Member of the **FSE/ToSC 2023/2024**

Editorial Board: **FSE/ToSC 2022/2023**

FSE/ToSC 2021/2022

*SPRING 2026:*³ Co-organizer (together with Prof. Marco Pedicini)

*ALPSY 2024:*⁴ Co-organizer (together with Prof. Arnab Roy, Prof. Christian Rechberger, and Reinhard Lueftenegger)

External Reviewer: Top-level conferences and journals, including: CRYPTO, EUROCRYPT, ASIACRYPT, FSE/ToSC, Journal of Cryptology, and DCC (Designs, Codes and Cryptography)

Publications

Please, note (1st) that IACR’s CRYPTO, EUROCRYPT and ASIACRYPT are generally considered the most important publication venues in the world for cryptographic academic research (A*-rank conference are highlighted), and (2nd)

¹<https://esp.ethereum.foundation/academic-grants>

²<https://casa.rub.de/en/jobs/casa-jump-start-program-for-postdocs>

³<https://lorenzo3grassi.github.io/SPRING26/spring26.html>

⁴<https://alpsy-workshop.github.io/alpsy24/>

that, in cryptography, authors are always listed in alphabetical order.

- TCHES 2025(4) B. Balon, L. Grassi, P. Méaux, T. Moos, F.-X. Standaert, M. J. Steiner: *“mid-pSquare: Leveraging the Strong Side-Channel Security of Prime-Field Masking in Software”*. DOI: 10.46586/TCHES.V2025.I4.486-519
- Designs, Codes and Cryptography, 2025 L. Grassi: *“On Generalizations of the Lai-Massey Scheme”*. DOI: 10.1007/S10623-025-01672-2
- FSE/ToSC 2025(2) L. Grassi, K. Koschatko, C. Rechberger: *“Poseidon and Neptune: Gröbner Basis Cryptanalysis Exploiting Subspace Trails”*. DOI: 10.46586/TOSC.V2025.i2.34-86
- TCHES 2025(2) C. Bouvier, L. Grassi, D. Khovratovich, K. Koschatko, C. Rechberger, F. Schmid, M. Schofnegger: *“Skyscraper: Fast Hashing on Big Primes”*. DOI: 10.46586/TCHES.V2025.I2.743-780
- FSE/ToSC 2024(4) F. Liu, K. Koschatko, L. Grassi, H. Yan, S. Chen, S. Banik, W. Meier : *“Opening the Blackbox: Collision Attacks on Round-Reduced Tip5, Tip4, Tip4’ and Monolith”*. DOI: 10.46586/TOSC.V2024.I4.97-137
- ASIACRYPT 2024 A. F. Gutierrez, L. Grassi, G. Leander, F. Sibleyras, Y. Todo: *“Generic Practical Cryptanalysis for the Sum of Round-Reduced Block Ciphers and ZIP-AES”*. DOI: 10.1007/978-981-96-0947-5_10
- FSE/ToSC 2024(3) L. Grassi, D. Khovratovich, R. Lüftenegger, C. Rechberger, M. Schofnegger, R. Walch: *“Monolith: Circuit-Friendly Hash Functions with New Nonlinear Layers for Fast and Constant-Time Implementations”*. DOI: 10.46586/tosc.v2024.i3.44-83
- SAC 2024 L. Grassi, F. Liu, C. Rechberger, F. Schmid, R. Walch, Q. Wang: *“Minimize the Randomness in Rasta-Like Designs: How Far Can We Go? (Application to Pasta)”*. DOI: 10.1007/978-3-031-82841-6_9
- EUROCRYPT 2024** L. Grassi, L. Masure, P. Méaux, T. Moos, F.-X. Standaert: *“Generalized Feistel Ciphers for Efficient Prime Field Masking”*. DOI: 10.1007/978-3-031-58734-4_7
- CRYPTO 2023** L. Grassi, Y. Hao, C. Rechberger, M. Schofnegger, R. Walch, Q. Wang: *“Horst Meets Fluid-SPN: Griffin for Zero-Knowledge Applications”*. DOI: 10.1007/978-3-031-38548-3_19
- CRYPTO 2023** L. Grassi, I. Manterola Ayala, M. Norberg Hovd, H. Raddum, Q. Wang, M. Øygarden: *“Cryptanalysis of Symmetric Primitives over Rings and a Key Recovery Attack on Rubato”*. DOI: 10.1007/978-3-031-38548-3_11
- CRYPTO 2023** F. Liu, L. Grassi, C. Bouvier, W. Meier, T. Isobe: *“Coefficient Grouping for Complex Affine Layers”*. DOI: 10.1007/978-3-031-38548-3_18
- FSE/ToSC 2023(2) L. Grassi: *“Bounded Surjective Quadratic Functions over $\mathbb{F}_p^n$ for MPC-/ZK-/FHE-Friendly Symmetric Primitives”*. DOI: 10.46586/tosc.v2023.i2.94-131
- AFRICACRYPT 2023 L. Grassi, D. Khovratovich, M. Schofnegger: *“Poseidon2: A Faster Version of the Poseidon Hash Function”*. DOI: 10.1007/978-3-031-37679-5_8
- AFRICACRYPT 2023 G. Giordani, L. Grassi, S. Onofri, M. Pedicini: *“Invertible Quadratic Non-Linear Functions over $\mathbb{F}_p^n$ via Multiple Local Maps”*. DOI: 10.1007/978-3-031-37679-5_7
- TCHES 2023(3) C. Dobraunig, L. Grassi, L. Helming, C. Rechberger, M. Schofnegger, R. Walch: *“Pasta: A Case for Hybrid Homomorphic Encryption”*. DOI: 10.46586/tches.v2023.i3.30-73
- EUROCRYPT 2023** L. Grassi, M. Øygarden, M. Schofnegger, R. Walch: *“From Farfalle to Megafono via Ciminion: The PRF Hydra for MPC Applications”*. DOI: 10.1007/978-3-031-30634-1_9
- ASIACRYPT 2022 L. Grassi, B. Mennink: *“Security of Truncated Permutation Without Initial Value”*. DOI: 10.1007/978-3-031-22966-4_21

- ACM CCS 2022 L. Grassi, D. Khovratovich, R. Lüftenegger, C. Rechberger, M. Schofnegger, R. Walch: *"Reinforced Concrete: Fast Hash Function for Zero Knowledge Proofs and Verifiable Computation"*. DOI: 10.1145/3548606.3560686
- FSE/ToSC 2022(3) L. Grassi, S. Onofri, M. Pedicini, L. Sozzi: *"Invertible Quadratic Non-Linear Layers for MPC-/FHE-/ZK-Friendly Schemes over $\mathbb{F}_p^n$ "*. DOI: 10.46586/tosc.v2022.i3.20-72
- FSE/ToSC 2022(2) A. Mehrdad, J. Daemen, L. Grassi, S. Mella: *"Differential Trail Search in Cryptographic Primitives with Big-Circle Chi"*. DOI: 10.46586/tosc.v2022.i2.253-288
- ACISP 2022 L. Grassi, C. Rechberger: *"Truncated Differential Properties of the Diagonal Set of Inputs for 5-round AES"*. DOI: 10.1007/978-3-031-22301-3_2
- FSE/ToSC 2022(1) L. Grassi, D. Khovratovich, S. Rønjom, M. Schofnegger: *"The Legendre Symbol and the Modulo-2 Operator in Symmetric Schemes over $(\mathbb{F}_p)^n$ "*. DOI: 10.46586/tosc.v2022.i1.5-37
- FSE/ToSC 2022(1) C. Cid, L. Grassi, A. Gunesing, R. Lüftenegger, C. Rechberger, M. Schofnegger: *"Influence of the Linear Layer on the Algebraic Degree in SP-Networks"*. DOI: 10.46586/tosc.v2022.i1.110-137
- FSE/ToSC 2021(2) L. Grassi, C. Rechberger, M. Schofnegger: *"Proving Resistance Against Infinitely Long Subspace Trails: How to Choose the Linear Layer"*. DOI: 10.46586/tosc.v2021.i2.314-352
- EUROCRYPT 2021 C. Dobraunig, L. Grassi, A. Guinet, D. Kuijsters: *"Ciminion: Symmetric Encryption Based on Toffoli-Gates over Large Finite Fields"*. DOI: 10.1007/978-3-030-77886-6_1
- USENIX-Security 2021 L. Grassi, D. Khovratovich, C. Rechberger, A. Roy, M. Schofnegger: *"Poseidon: A New Hash Function for Zero-Knowledge Proof Systems"*. Link: <https://www.usenix.org/conference/usenixsecurity21/presentation/grassi>
- INDOCRYPT 2020 L. Grassi, M. Schofnegger: *"Mixture Integral Attacks on Reduced-Round AES with a Known/Secret S-Box"*. DOI: 10.1007/978-3-030-65277-7_14
- ASIACRYPT 2020 M. Eichlseder, L. Grassi, R. Lüftenegger, M. Øyegarden, C. Rechberger, M. Schofnegger, Q. Wang: *"An Algebraic Attack on Ciphers with Low-Degree Round Functions: Application to Full MiMC"*. DOI: 10.1007/978-3-030-64837-4_16
- SAC 2020 T. Cui, L. Grassi: *"Algebraic Key-Recovery Attacks on Reduced-Round Xoofff"*. DOI: 10.1007/978-3-030-81652-0_7
- SAC 2020 L. Grassi, G. Leander, C. Rechberger, C. Tezcan, F. Wiemer: *"Weak-Key Distinguishers for AES"*. DOI: 10.1007/978-3-030-81652-0_6
- EUROCRYPT 2020 L. Grassi, R. Lüftenegger, C. Rechberger, D. Rotaru, M. Schofnegger: *"On a Generalization of Substitution-Permutation Networks: The HADES Design Strategy"*. DOI: 10.1007/978-3-030-45724-2_23
- Designs, Codes and Cryptography, 2020 L. Grassi, C. Rechberger: *"Revisiting Gilbert's known-key distinguisher"*. DOI: 10.1007/s10623-020-00756-5
- ASIACRYPT 2019 M. Albrecht, C. Cid, L. Grassi, D. Khovratovich, R. Lüftenegger, C. Rechberger, M. Schofnegger: *"Algebraic Cryptanalysis of STARK-Friendly Designs: Application to MARVELlous and MiMC"*. DOI: 10.1007/978-3-030-34618-8_13
- SAC 2019 L. Grassi: *"Probabilistic Mixture Differential Cryptanalysis on round-reduced AES"*. DOI: 10.1007/978-3-030-38471-5_3
- ESORICS 2019 M. Albrecht, L. Grassi, L. Perrin, S. Ramacher, C. Rechberger, D. Rotaru, A. Roy, M. Schofnegger: *"Feistel Structures for MPC, and More"*. DOI: 10.1007/978-3-030-29962-0_8
- FSE/ToSC 2018(2) L. Grassi: *"Mixture Differential Cryptanalysis: a New Approach to Distinguishers and Attacks on round-reduced AES"*. DOI: 10.13154/tosc.v2018.i2.133-160

- ASIACRYPT 2018 L. Grassi, M. Naya-Plasencia, A. Schrottenloher: “Quantum Algorithms for the k -xor Problem”. DOI: 10.1007/978-3-030-03326-2_18
- CRYPTO 2018 C. Dobraunig, M. Eichlseder, L. Grassi, V. Lallemand, G. Leander, E. List, F. Mendel, C. Rechberger: “Rasta: A Cipher with Low ANDdepth and Few ANDs per Bit”. DOI: 10.1007/978-3-319-96884-1_22
- CT-RSA 2018 L. Grassi: “MixColumns Properties and Attacks on (Round-Reduced) AES with a Single Secret S -Box”. DOI: 10.1007/978-3-319-76953-0_13
- CT-RSA 2018 Q. Wang, L. Grassi, C. Rechberger: “Zero-Sum Partitions of PHOTON Permutations”. DOI: 10.1007/978-3-319-76953-0_15
- EUROCRYPT 2017 L. Grassi, C. Rechberger, S. Rønjom: “A New Structural-Differential Property of 5-Round AES”. DOI: 10.1007/978-3-319-56614-6_106
- FSE/ToSC 2016(2) L. Grassi, C. Rechberger, S. Rønjom: “Subspace Trail Cryptanalysis and its Applications to AES”. DOI: 10.13154/tosc.v2016.i2.192-225
- ASIACRYPT 2016 M.R. Albrecht, L. Grassi, C. Rechberger, A. Roy, T. Tiessen: “MiMC: Efficient Encryption and Cryptographic Hashing with Minimal Multiplicative Complexity”. DOI: 10.1007/978-3-662-53887-6_7
- ACM CCS 2016 L. Grassi, C. Rechberger, D. Rotaru, P. Scholl, N. P. Smart: “MPC-Friendly Symmetric Key Primitives”. DOI: 10.1145/2976749.2978332
- INDOCRYPT 2016 L. Grassi, C. Rechberger: “Practical Low Data-Complexity Subspace-Trail Cryptanalysis of Round-Reduced PRINCE”. DOI: 10.1007/978-3-319-49890-4_18
- SPACE 2015 G. M. Bertoni, L. Grassi, F. Melzani: “Simulations of Optical Emissions for Attacking AES and Masked AES”. DOI: 10.1007/978-3-319-24126-5_11
- M. Sansottera, L. Grassi and A. Giorgilli: “On the relativistic Lagrange-Laplace secular dynamics for extrasolar systems.” Proceedings of the International Astronomical Union, **9**, 2014. DOI: 10.1017/S174392131400787X

Teaching Activities and Student Supervision

(WiSe $\equiv$ Winter Semester – SuSe $\equiv$ Summer Semester)

- WiSe 2025/2026 "Introduction to Cryptology", Bachelor's course (TU/e), Lecturer
- WiSe 2023/2024 "Symmetric Cryptanalysis", Master's course (RUB), Teaching Assistant
- WiSe 2022/2023 "Cryptology", Master's course (Radboud University), Guest Lecturer (1 lecture)
- WiSe 2022/2023 "Computer Security", Bachelor's course (Radboud University), Teaching Assistant
- WiSe 2019/2020 – "Applied Cryptography 1", Master's course (TU Graz), Co-Lecturer
2018/2019 –
2017/2018
- SuSe 2018/2019 – "Applied Cryptography 2", Master's course (TU Graz), Co-Lecturer
2017/2018 –
2016/2017

Supervision of Bachelor/Master Thesis/Project

(Co-supervised with Prof. Bart Mennink* and Prof. Christian Rechberger[†])

- Quinn Ketelaars Master Thesis 2025/2026: “Round-Level Gröbner Basis Attacks of Poseidon and Neptune”
- Sjors Clabbers* Bachelor Thesis 2021/2022: “Linear Cryptanalysis of round-reduced GIFT”
- Giulio Arrigoni[†] Bachelor Thesis 2019/2020: “Analysis of 5- and 6-round AES”
- Bernhard Ablinger[†] Master Project 2018/2019: “Higher-Order Differential over $GF(p)$ ”
- Matthias Niel[†] Master Project 2017/2018: “Mean and Variance Distinguishers for AES”

Talks

- 11/2025 **Crypto Working Group**: “Symmetric Primitives for MPC-/ZK-/HE-Applications” Utrecht (Netherlands), joint work
- 12/2024 **University of Novi Sad**: “Symmetric Primitives for Future Applications: Poseidon” virtual
- 06/2024 **Xuejia Lai’s 70th Birthday**: “Lai-Massey Scheme: Possible Generalizations” virtual
- 03/2024 **FSE/ToSC**: “Bounded Surjective Quadratic Functions over $\mathbb{F}_p^n$ for MPC-/ZK-/FHE-Friendly Symmetric Primitives” Leuven (Belgium)
- 12/2023 **Asian Workshop on Symmetric Key Cryptography (ASK)**: “Horst and Amaryllises: Possible Generalizations of the Feistel and of the Lai-Massey Schemes” Guangzhou (China)
- 09/2023 **Convegno UMI Crittografia e Codici**: “Symmetric Primitives for MPC-/ZK-/FHE-Applications” Perugia (Italy), joint work
- 09/2023 **Università degli Studi Roma Tre**: “Symmetric Primitives for MPC-/ZK-/FHE-Applications” Roma (Italy), joint work
- 03/2023 **FSE/ToSC**: “Security of Truncated Permutation Without Initial Value” Kobe (Japan), joint work with: S. Onofri, M. Pedicini, L. Sozzi
- 12/2022 **ASIACRYPT**: “Security of Truncated Permutation Without Initial Value” Taipei (Taiwan), joint work with: B. Mennink
- 11/2022 **ACISP**: “Truncated Differential Properties of the Diagonal Set of Inputs for 5-round AES” virtual, joint work with: C. Rechberger
- 09/2022 **FrisiaCrypt**: “Re-thinking Feistel and Lai-Massey schemes: the birth of Horst and Amaryllises” Terschelling (the Netherlands)
- 09/2022 **ERC Workshop: New Directions in Cryptanalysis of AES**: “Cryptanalysis of round-reduced AES” Tel Aviv (Israel), joint work
- 07/2022 **Université Catholique de Louvain**: “Design and Security Analysis of Symmetric Primitives for Multi-Party Computation” Louvain (Belgium), joint work
- 05/2022 **University of Twente**: “Symmetric Primitives for Multi-Party Computation (MPC) Application” Twente (the Netherlands), joint work
- 03/2022 **FSE/ToSC**: “The Legendre Symbol and the Modulo-2 Operator in Symmetric Schemes over $(\mathbb{F}_p)^n$ ” Athens (Greece), joint work with: D. Khovratovich, S. Rønjom, M. Schofnegger
- 03/2022 **Universität Innsbruck**: “Symmetric Primitives for Multi-Party Computation (MPC) Application” Innsbruck (Austria), joint work
- 12/2020 **INDOCRYPT**: “Mixture Integral Attacks on Reduced-Round AES with a Known/Secret S-Box”, virtual, joint work with: M. Schofnegger
- 12/2020 **De Componendis Cifris**: “Symmetric-Key Encryption Schemes for MPC Applications”, virtual, joint work
- 11/2020 **Università degli Studi di Milano**: “Cryptanalysis of round-reduced AES: New Results”, virtual, joint work
- 10/2020 **SAC**: “Weak-Key Distinguishers for AES”, virtual, joint work with: G. Leander, C. Rechberger, C. Tezcan, F. Wiemer
- 01/2020 **Dagstuhl Seminar**: “Higher-Order Differential Attacks on Ciphers with Low-Degree Polynomial S-Boxes in $GF(2^n)$: Open Problems”, Dagstuhl (Germany), joint work
- 11/2019 **Università degli Studi Roma Tre**: “Cryptanalysis of round-reduced AES: New Results”, Roma (Italy), joint work

- 08/2019 **SAC**: “Probabilistic Mixture Differential Cryptanalysis on round-reduced AES”, Waterloo (Canada)
- 09/2018 **Centre INRIA de Paris**: “Cryptanalysis of round-reduced AES: New Results”, Paris (France), joint work
- 04/2018 **CT-RSA**: “MixColumns Properties and Attacks on (Round-Reduced) AES with a Single Secret S-Box”, San Francisco (USA)
- 04/2018 **CT-RSA**: “Zero-Sum Partitions of PHOTON Permutations”, San Francisco (USA), joint work with: Q. Wang, C. Rechberger
- 03/2018 **FSE/ToSC**: “Mixture Differential Cryptanalysis: a New Approach to Distinguishers and Attacks on round-reduced AES”, Paris (France)
- 11/2017 **Technical University of Denmark**: “Cryptanalysis of AES: New Results”, Copenhagen (Denmark), joint work
- 05/2017 **EUROCRYPT**: “A New Structural-Differential Property of 5-Round AES”, Paris (France), joint work with: C. Rechberger, S. Rønjom
- 03/2017 **FSE/ToSC**: “Subspace Trail Cryptanalysis and its Applications to AES”, Tokyo (Japan), joint work with: C. Rechberger, S. Rønjom
- 12/2016 **INDOCRYPT**: “Practical Low Data-Complexity Subspace-Trail Cryptanalysis of Round-Reduced PRINCE”, Kolkata (India), joint work with: C. Rechberger
- 12/2015 **SPACE**: “Simulations of Optical Emissions for Attacking AES and Masked AES”, Jaipur (India), joint work with: G. M. Bertoni, F. Melzani

References

Prof. **Tanja Lange**

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Prof. **Christian Rechberger**

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